The Fundamental Theorem of Calculus

ANSWER KEY

Part 2: evaluating definite integrals

1 Evaluate each definite integral:

a $\int_1^3 (2x+1)\,dx$ $[x^2+x]_1^3 = 12 - 2 = 10.$

b $\int_0^{\pi} \sin x\,dx$ $[-\cos x]_0^{\pi} = 1 - (-1) = 2.$

c $\int_1^4 \sqrt{x}\,dx$ $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}.$

2 Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.
$\left[\frac{1}{2}e^{2x}\right]_0^1 = \frac{1}{2}(e^2 - 1) \approx 3.195.$

Part 1: differentiating an integral

3 Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt$.
$= x^3 + 1$ (directly, by FTC Part 1).

4 Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.
$= \sin(x^2) \cdot 2x = 2x\sin(x^2).$

5 Let $g(x) = \int_0^x \sqrt{4+t^2}\,dt$. Find $g'(3)$.
$g'(x) = \sqrt{4+x^2}$, so $g'(3) = \sqrt{13}$.

Average value and applications

6 Find the average value of $f(x) = x^2$ on $[0, 3]$.
$\frac{1}{3}\int_0^3 x^2\,dx = \frac{1}{3} \cdot 9 = 3.$

7 A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.
Distance $= \int_0^4 (3t^2+2)\,dt = [t^3+2t]_0^4 = 64 + 8 = 72$ m. Average velocity $= \frac{72}{4} = 18$ m/s.

Further FTC practice

8 Evaluate $\int_0^2 (x^3 - 2x + 3)\,dx$.
$\left[\frac{x^4}{4} - x^2 + 3x\right]_0^2 = (4 - 4 + 6) - 0 = 6.$

9 Find $\frac{d}{dx}\int_x^5 \cos(t^2)\,dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.
$\int_x^5 \cos(t^2)\,dt = -\int_5^x \cos(t^2)\,dt$, so the derivative is $-\cos(x^2)$.

10 Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.
$\frac{1}{\pi}\int_0^{\pi} \sin x \, dx = \frac{1}{\pi}[-\cos x]_0^{\pi} = \frac{2}{\pi} \approx 0.637$.

Mean Value Theorem for integrals

11 Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).
Average value $= \frac{1}{3}\int_0^3 x^2 \, dx = 3$ (found earlier). Solve $c^2 = 3$: $c = \sqrt{3}$ (taking the positive root within $[0, 3]$).

12 Evaluate $\int_2^5 f(x) \, dx$ given that the average value of f on $[2, 5]$ is 7.
Average value $= \frac{1}{5-2}\int_2^5 f(x) \, dx = 7$, so $\int_2^5 f(x) \, dx = 7(3) = 21$.

Integrals of piecewise functions

13 Evaluate $\int_0^4 f(x) \, dx$ where $f(x) = x$ for $x \le 2$ and $f(x) = 2x - 2$ for $x > 2$.
$\int_0^2 x \, dx + \int_2^4 (2x - 2) \, dx = \left[\frac{x^2}{2}\right]_0^2 + [x^2 - 2x]_2^4 = 2 + (8 - 0) = 10$.